

Matched pretrained comparison: result

Our incumbent PET vs. Gregor's complete pretrained PET2-small

PET configuration-comparison lane

September 21, 2026

Result

No adequate arm.

Neither configuration cleared the predeclared safeguards, so this comparison recommends neither. That is a result, not a failure to produce one.

Matched paired effect, ours – theirs, on the predeclared seven-bin E_{avail} recovery:

$$\bar{d} = -0.1129, \quad 95\% \text{ CI} = [-0.1471, -0.0786], \quad n = 8 \text{ pairs.}$$

Non-inferiority margin $\delta = 0.02$, switching threshold $\delta_{\text{switch}} = 0.04$, both frozen before any comparative result existed.

The interval is wider than δ (0.0343 against 0.02). This comparison resolves the DIRECTION of the difference, not non-inferiority at the frozen margin; the margin was not widened to change that.

The pilot observations are **excluded**: the pilot chooses n . Reusing its observations in the interval that n was chosen for would make the interval conditional on its own width.

What this measures, and what it cannot

The arms differ at step 1 only

the arms DIFFER: ours is the production cluster cloud, theirs is the complete PET2-small arm over typed objects

Step 2 is identical

the arms are IDENTICAL: the production PET on the production truth cloud, same weights
initialisation policy, same recipe

The limit this imposes

his configuration's behaviour on the truth side, because that configuration does not exist – his vocabulary has no truth analogue. Any recommendation states this limit in the conclusion, not a footnote

Absolute adequacy, asked of each arm alone

arm	mean recovery	floor	adequate
ours	0.3037	0.5560	no
theirs	0.4165	0.5560	no

The floor is $0.8 \times$ the applicable reference (0.6950).

An arm failing this is ineligible, and one arm's failure does not disqualify the other: eligibility is asked of each arm on its own.

Regional safeguard, on the reporting cells

region	ours	theirs	floor
low_acceptance	0.0558	0.1526	0.0084
moderate	0.2264	0.4142	0.4659
good	0.5381	0.6886	0.5853

Regions are defined on the $(p_T, p_{||})$ reporting cells, never by stratifying the seven marginal bins. Each floor is $0.6\times$ *that region's own* reference.

Safeguard coverage: events whose $(p_T, p_{||})$ falls off the reporting grid are in the aggregate score but in no region — 0.00% of truth mass.

Low-acceptance events: retained, reported separately

truth mass fraction	0.3103
injected displacement share	0.2657
mean recovery, ours	0.0558
mean recovery, theirs	0.1526

Scope of any recommendation here

These events are retained and reported. The acceptance reference is not an impossibility bound, and this band is not called fundamentally unresolvable. Any recommendation states its limits here.

Stability across seeds

seed	$d = \text{ours} - \text{theirs}$
127	-0.1679
139	-0.1654
151	-0.0960
163	-0.1113
179	-0.0534
191	-0.1363
211	-0.0760
223	-0.0967
mean	-0.1129
s.d.	0.0410

Component comparison: evidence and preference kept apart

The twelve pinned categories, read from CONFIGURATION_COMPARISON-20260918.md

#	category	recommendation
1	Object membership, filtering, overlap, primary lepton	Borrow the typed-object vocabulary as a candidate, do not borrow the muon-as-token choice. Keep the muon distinguished.
2	Token features, PID/type encoding, missingness, padding	Borrow the richer per-token feature set and the learned type embedding. Do not borrow zero-fill-for-absent, and do not reuse index 0 for a real class. Reserve index 0 for padding and shift real codes by one.
3	Coordinate transforms and normalization	Keep ours. Recommend to Gregor that φ be encoded periodically — this is the one place where we have already paid for the lesson. Do not adopt input-side LayerNorm.
4	Event globals	Borrow the per-PID / per-family log energy sums, computed over the <i>whole</i> event before any cap. They are a fixed-width, cap-proof channel and cost essentially nothing. This is the cheapest candidate on the list. Do not borrow the $\gamma\gamma$-mass-or-zero encoding without a companion validity flag.
5	Pooling versus individual-object attention	Keep pooling as the practical default; his individual routing is not shown to be better and is measurably more expensive at our multiplicities. This is a development choice under an inconclusive result, not a finding.
6	Caps, ordering, overflow aggregation, merged counts	Borrow the aggregate-overflow idea (we independently built the same thing as arm D) and test it before anything else. Do not borrow his ordering, his default-off setting, or his unweighted auxiliary mean — use energy-ranked retention (ours) and an energy-weighted merge.
7	Architecture, capacity, attention structure	Keep our architecture and its size. Treat capacity as an open question, not a settled one. A $19\times$ parameter gap is large enough that "we are under-parameterized" is a live hypothesis, and it is cheap to test one width step.
8	Initialization and pretraining	Keep random init with the frozen seed. Ask Gregor directly for the checkpoint files and their hashes before planning any pretrained arm — do not plan around a URL.
9	Training objective, optimizer, regularization, stopping	Borrow warmup + cosine decay and gradient clipping; do not borrow weight decay until someone measures its effect on the reweighting bias. Both are candidates, neither is adopted.
10	OmniFold integration, ratio conventions, validation	Keep ours in full. If a typed-object representation is ever adopted, it must enter through our dump and loader, carrying weights and flags — not by importing his .pb files.
11	Training and inference cost	Order candidate changes by campaign cost, not by per-train cost.
12	Numerical behaviour	Keep ours.

Provenance

commit	68cf9d29f8ab
campaign directory	/pscratch/sd/j/josephrb/campaign-20260920
closure inputs	fa6b346316024216
truth rows scored	—
aggregate reference	0.6950

Scope

What this is not

PET remains diagnostic method development. This deck is **not** a publication adoption, a covariance construction, a systematic, a central-value change, or a Gate-6 action, and it discharges no open item.

Three differences from Gregor's configuration are **declared, not eliminated**: muon presence via `muon_E > 0` rather than the MINOS match, `prong_dEdXMean` substituted for the absent `prong_part_dEdXMean`, and a proportional cap split that is ours rather than his. Any reading of the result carries them.